

MATHEMATICS (SINDH BOARD)

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CLASS

11th

CHAPTER

6

EXERCISE

6.4

Q1: Two fair coins are tossed. Find the Probability of

Getting:

- i. Same faces ii. All heads iii. At most one head.
iv. At least two tail v. At least one tail.

i. Same faces: $S = \{HH, HT, TH, TT\}$

$$A = \{HH, TT\}$$

$$O(A) = 2$$

$$P(E) = \frac{O(E)}{O(S)}$$

There are two possible outcomes of same faces if two fair coin are tossed.

Therefore, $O(S) = \{HH, HT, TH, TT\}$

$$O(A) = \{HH, TT\}$$

$$O(S) = 4$$

$$O(A) = 2$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{2}{4} = \frac{1}{2}$$

ii. All Heads.

$$S = \{HH, HT, TH, TT\}$$

$$P(A) = \frac{O(A)}{O(S)}$$

$$O(A) = 1, \quad O(S) = 4$$

There are one possible outcome of all heads if two fair coins are tossed.

$$P(A) = \frac{1}{4}$$

Q#2: Two dice, one red and the other green, are rolled simultaneously. The numbers of dots on the tops are added. Find the Probability of getting a sum of:
 i- 8 ii- 10 iii- 12

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Solution:
 Two dice are rolled then the sample points are:

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

$$O(S) = 6 \times 6 = 36$$

i- 8

Solution:

$$A = \{(2,6), (3,5), (4,4), (5,3), (6,2)\} \quad \therefore \begin{matrix} O(A) = 5 \\ O(S) = 36 \end{matrix}$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{5}{36}$$

ii- 10

Solution:

$$B = \{(4,6), (5,5), (6,4)\}$$

$$\therefore \begin{matrix} O(B) = 3 \\ O(S) = 36 \end{matrix}$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{3}{36} = \frac{1}{12}$$

iii- 12

Solution

$$C = \{(6,6)\}$$

$$\therefore \begin{matrix} O(C) = 1 \\ O(S) = 36 \end{matrix}$$

$$P(C) = \frac{O(C)}{O(S)} = \frac{1}{36}$$

Q#3: A bag contains 40 balls out of which 5 are green, 15 are red and the remaining are black. A ball is drawn out of the bag. Find the Probability of getting:

i. The ball is green

ii. The ball is black

iii. The ball is not green.

Solution:

Green balls are 5

Red balls are 15

Black balls are 20.

i. The ball is green:

$$P(G) = \frac{O(G)}{O(S)} = \frac{5}{40}$$

$$P(G) = \frac{1}{8}$$

$$O(S) = 40$$

$$O(G) = 5$$

$$O(R) = 15$$

$$O(B) = 20$$

ii. The ball is black.

$$P(B) = \frac{O(B)}{O(S)} = \frac{20}{40} = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

iii. The ball is not green:

$$P(\text{not green}) = \frac{P(R) + P(B)}{O(S)} = \frac{15 + 20}{40}$$

$$P(\text{not green}) = \frac{35}{40} = \frac{7}{8} \quad \underline{\text{Ans}}$$

Q#4: If drawn card from a well shuffled deck of cards, find the Probability of:

- i. getting a king
- ii. getting a club.
- iii. getting a face card.

Solution:

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There are 52 cards are in deck of cards.

Now, find the probability of:

- i. getting a King

Solution:

Total King in deck of cards = 4

$$O(K) = 4$$

Total cards = $O(S) = 52$

$$P(K) = \frac{O(K)}{O(S)} = \frac{4}{52} = \frac{1}{13}$$

$$P(K) = \frac{1}{13}$$

- ii. getting a club:

Solution:

Total clubs in deck of cards = 13

$$O(C) = 13$$

Total cards = 52

$$P(C) = \frac{O(C)}{O(S)} = \frac{13}{52} = \frac{1}{4}$$

- iii. getting a face card:

Solution:

Total face cards in deck of cards = 12

$$O(F) = 12$$

Total cards = $O(S) = 52$

$$P(F) = \frac{O(F)}{O(S)} = \frac{12}{52} = \frac{3}{13}$$

Q#5: A die is rolled. Find the probability using Venn diagram of getting:

- i. an even number
- ii. a number greater than 4.
- iii. a number which is even and greater than 4.

Solution:

$$S = \{1, 2, 3, 4, 5, 6\}$$

i. an even number:

Even numbers = 2, 4, 6

$$O(A) = 3$$

$$\text{Total numbers} = O(S) = 6$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{3}{6} = \frac{1}{2}$$

ii. a number greater than 4:

a number greater than 4 = 5, 6

$$O(B) = 2$$

$$\text{Total number} = O(S) = 6$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{2}{6} = \frac{1}{3}$$

iii. a number which is even and greater than 4:

a number which is even and greater than 4 = 6

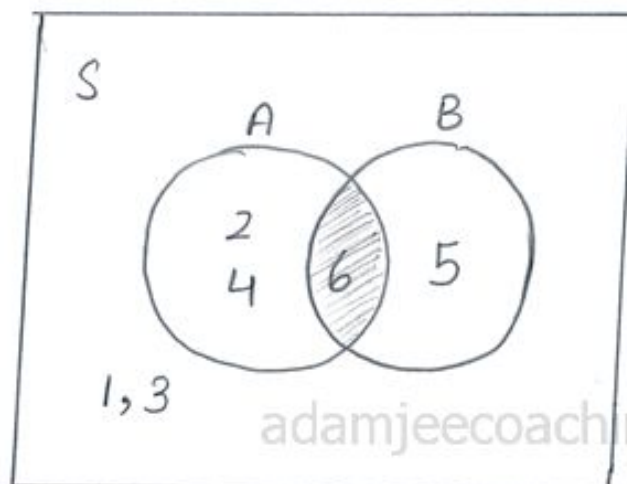
$$O(C) = 1, \text{ Total number} = O(S) = 6$$

$$P(C) = \frac{O(C)}{O(S)} = \frac{1}{6}$$

$$A \cap B = \{2, 4, 6\} \cap$$

$$\{5, 6\}$$

$$A \cap B = \{6\}$$

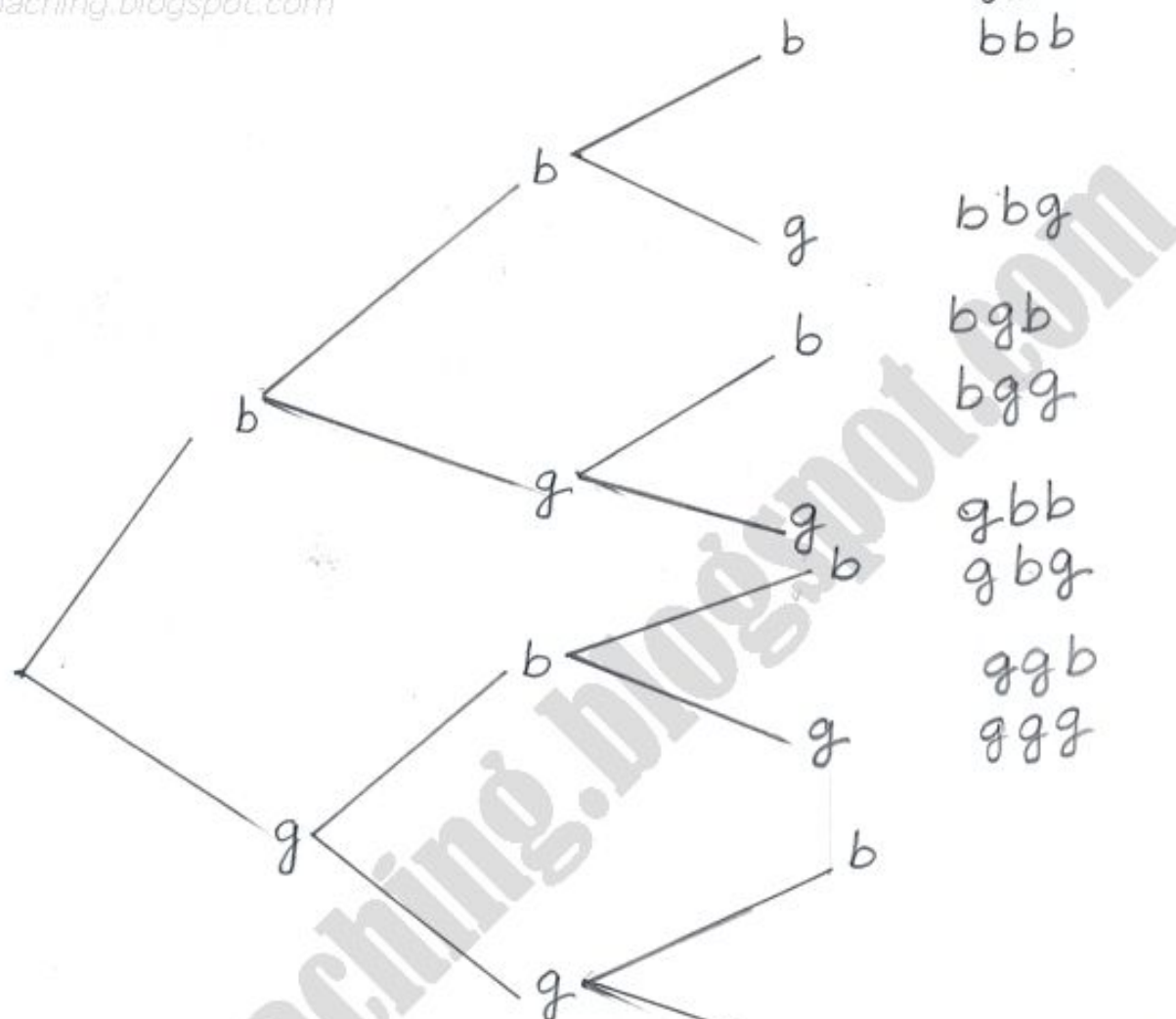


Q#6: In a three child family, by using tree diagram. Find the probability of having:

- i. three boys ii. exactly two boys iii. at least one girl.

Solution:

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$S = \{ bbb, bbg, bbg, bgg, gbb, gbg, ggb, ggg \}$

$n(S) = 8$

i. three boys.

$n(\text{three boys}) = bbb$

$n(A) = 1$

$P(\text{three boys}) = \frac{n(A)}{n(S)} = \frac{1}{8}$

iii. at least one girl:

$n(\text{at least one girl}) = bbg, bbg, bgg, gbb, gbg, ggb, ggg$

$n(C) = 7$

$n(S) = 8$

$P(\text{at least one girl}) = \frac{n(C)}{n(S)}$

$P(C) = \frac{7}{8}$

ii. exactly two boys:

$n(\text{exactly two boys}) = \{ bbg, bbg, gbb \}$

$n(B) = 3$

$P(\text{exactly two boys}) = \frac{3}{8}$

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Q#7: In a single throw of two fair dice, find the Probability that the Product of the numbers on the dice is :
i- between 2 and 10 (both inclusive) ii- divisible by 5.

Solution:

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$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), \\ (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), \\ (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), \\ (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

i- between 2 and 10: (both inclusive) :

$$A = \{(1,2), (1,3), (1,4), (1,5), (1,6), (2,1), \\ (2,2), (2,3), (2,4), (2,5), (3,1), (3,2), \\ (3,3), (4,1), (4,2), (5,1), (5,2), (6,1)\}$$

$$O(A) = 18$$

$$O(S) = 36$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{18}{36} = \frac{1}{2}$$

ii- divisible by 5:

$$B = \{(1,5), (2,5), (3,5), (4,5), (5,1), (5,2), \\ (5,3), (5,4), (5,5), (5,6), (6,5)\}$$

$$O(B) = 11$$

$$O(S) = 36$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{11}{36} \quad \underline{\underline{\text{Ans}}}$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 3, 5\}$$

$$B = \{2, 4, 6\}$$

$$C = \{2, 3, 5\}$$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 3, 5\}$$

$$B = \{2, 4, 6\}$$

$$A \cup B = S$$

$$A \cap B = \emptyset$$

mutually Exclusive events.

$$S = \{\text{Head}, \text{Tail}\}$$

$$A = \{\text{Head}\}$$

$$B = \{\text{Tail}\}$$

$$A \cup B = S$$

Q#8: A marble is drawn at random from a box containing 20 red, 10 white, 25 orange and 15 blue marbles.

Find the probability that it is:

- i. Orange or red ii. not blue or red
iii. red, white or blue.

Solution:

Total red marbles = 20

Total white marbles = 10

Total orange marbles = 25

Total blue marbles = 15

Total marbles = $20 + 10 + 25 + 15$

$O(S) = 70$

i. Orange or red:

Orange or red = $25 + 20$

Total marbles = 70

$P(\text{orange or red}) = \frac{25 + 20}{70} = \frac{45}{70}$

$P(\text{orange or red}) = \frac{9}{14}$

ii. not blue or red:

not blue or red = $70 - 15 - 20$

not blue or red = 35

Total marbles = 70

$P(\text{not blue or red}) = \frac{35}{70}$

$P(\text{not blue or red}) = \frac{1}{2}$

iii. red, white or blue.

red, white or blue = $20 + 10 + 15$

Total marbles = 70

$P(\text{red, white or blue}) = \frac{20 + 10 + 15}{70}$

$P(\text{red, white or blue}) = \frac{45}{70}$

$P(\text{red, white or blue}) = \frac{9}{14}$ Ans

Q#9: The King, queen and jack of clubs are removed from a deck of 52 playing cards and then shuffled. A card is drawn from the remaining cards. Find the probability of getting:

i. a heart

ii. a queen

iii. a club

iv. '9' of red color.

Solution:

Total cards = 52

Now, we remove King, queen and jack of club then remaining cards are 49

$$52 - 3 = 49$$

$$O(S) = 49$$

i. a heart.

Total heart in deck of cards = 13

$$O(A) = 13$$

$$P(\text{heart}) = \frac{O(\text{heart})}{O(S)} = \frac{13}{49}$$

ii. a queen.

Total queens in deck of cards = 4
but we remove one queen in our deck of cards the
3 queens are remaining:

$$O(\text{Queen}) = 3$$

$$O(S) = 49$$

$$P(\text{queen}) = \frac{3}{49}$$

iii. a club.

Total club in deck of cards = 13
but we remove one King one queen
and jack of club then remaining
club are 10.

$$O(\text{club}) = 10$$

$$O(S) = 49$$

$$P(\text{club}) = \frac{10}{49}$$

IV- '4' of red colour:

Total nine of red color is 2

$$O(9 \text{ of red color}) = 2$$

$$O(S) = 49$$

$$P(9 \text{ of red color}) = \frac{2}{49} \quad \underline{\underline{\text{Ans}}}$$

Q#10: A pair of fair dice is thrown. If the two numbers appearing are different, find the probability that
i- the sum is 10, ii- the sum is six or less.

Solution:

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$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

$$O(S) = 36$$

i- The sum is 10.

$$A = \{(4,6), (5,5), (6,4)\}$$

$$O(A) = 2$$

$$O(S) = 36$$

$$\text{Probability} = \frac{\text{Favourable outcomes}}{\text{total outcomes.}}$$

$$= \frac{2}{36}$$

$$P(\text{The sum is 10}) = \frac{1}{18}$$

ii- The sum is 6 or less.

$$B = \{(1,2), (1,3), (1,4), (1,5), (2,1), (2,3), (2,4), (3,1), (3,2), (4,1), (4,2), (5,1)\}$$

$$O(B) = 12$$

$$O(S) = 36$$

$$P(\text{The sum is 6 or less}) = \frac{12}{36}$$

$$P(\text{The sum is 6 or less}) = \frac{1}{3}$$

Q#11: Given that $P(A) = 0.3$, $P(B) = 0.7$, $P(A \cap B) = 0.21$
then find:

i- $P(A/B)$ ii- $P(B/A)$

Solution:

$$P(A) = 0.3$$

$$P(B) = 0.7$$

$$P(A \cap B) = 0.21$$

Now find

i- $P(A/B)$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A/B) = \frac{0.21}{0.7}$$

$$P(A/B) = 0.3$$

$$\text{ii- } P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(B/A) = \frac{0.21}{0.3}$$

$$P(B/A) = 0.7 \quad \underline{\underline{\text{Ans}}}$$

Q#12: If one card is selected at random from a deck of 52 playing cards, what is probability that the card is a club or a face or both?

Solution:

Total cards = 52

$$O(S) = 52$$

Total club cards = 13

$$O(A) = 13$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{13}{52}$$

Total face cards = 12

$$O(B) = 12$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{12}{52}$$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52}$$

$$\therefore P(A \cup B) + P(A \cap B) = P(A) + P(B)$$

$$P(A \cap B) = \frac{3}{52}$$

$$P(A \cup B) = \frac{13+12-3}{52}$$

$$P(A \cup B) = \frac{22}{52}$$

$$P(A \cup B) = \frac{11}{26}$$

Q#13: From two events A and B, $P(A) = 0.5$, $P(B) = 0.2$, $P(A \cup B) = 0.4$, then find $P(A \cap B)$:

Solution:

Given that:

From two events A and B:

$$P(A) = 0.5, \quad P(B) = 0.2, \quad P(A \cup B) = 0.4,$$

$$P(A \cap B) = ?$$

Using Addition Law of Probability:

$$P(A \cup B) + P(A \cap B) = P(A) + P(B)$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$P(A \cap B) = 0.5 + 0.2 - 0.4$$

$$P(A \cap B) = 0.3$$

Ans

Q#13: A natural number is chosen out of first 35 natural numbers. What is the probability that the chosen number is divisible by 8 or 9?

Total natural numbers = $\{1, 2, 3, 4, 5, \dots, 35\}$

$$O(S) = 35$$

Let A and B are two events:

A is an event which are divisible by 8 or

B is an event which are divisible by 9.

$$A = \{8, 16, 24, 32\}$$

$$O(A) = 4$$

$$B = \{9, 18, 27\}$$

$$O(B) = 3$$

$$P(A) = \frac{O(A)}{O(S)} = \frac{4}{35}, \quad P(B) = \frac{O(B)}{O(S)} = \frac{3}{35}$$

$$P(\text{divisible by 8 or 9}) = \frac{4}{35} + \frac{3}{35} = \frac{7}{35}$$

$$P(\text{divisible by 8 or 9}) = \frac{1}{5} \quad \underline{\underline{\text{Ans}}}$$

Q#15: A bag contain 15 black, 25 red and 10 white balls. A ball is drawn at random. Find the Probability that it is either red or white?

Solution:

Black balls = 15

Red balls = 25

White Balls = 10

Total Balls = $15 + 25 + 10 = 50$

$$O(S) = 50$$

Red balls = $R = O(R) = 25$

White balls = $W = O(W) = 10$

$$P(R) = \frac{O(R)}{O(S)} = \frac{25}{50} = \frac{1}{2}$$

$$P(W) = \frac{O(W)}{O(S)} = \frac{10}{50} = \frac{1}{5}$$

$$P(\text{Balls are red or white}) = \frac{1}{2} + \frac{1}{5} = \frac{5+2}{10}$$

$$P(\text{Ball are red or white}) = \frac{7}{10} \quad \underline{\underline{\text{Ans}}}$$

Q#16: Two events A and B are such that $P(A) = \frac{1}{4}$, $P(A/B) = \frac{1}{2}$, $P(B/A) = \frac{2}{3}$. And $P(A \cap B)$ and $P(B)$?

Solution:

Given that:

$$P(A) = \frac{1}{4}$$

$$P(A/B) = \frac{1}{2}$$

$$P(B/A) = \frac{2}{3}$$

$$P(A \cap B) = ? , P(B) = ?$$

Now, Using formula of Conditional Probability

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$\frac{2}{3} = \frac{P(A \cap B)}{1/4}$$

$$\frac{2}{3} = 4(P(A \cap B))$$

$$\frac{2}{3 \times 4} = P(A \cap B)$$

$$\frac{1}{6} = P(A \cap B)$$

$$P(A \cap B) = \frac{1}{6}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B) = \frac{P(A \cap B)}{P(A/B)}$$

$$P(B) = \frac{1/6}{1/2}$$

$$P(B) = \frac{1}{6} \div \frac{1}{2}$$

$$P(B) = \frac{1}{6} \times \frac{2}{1}$$

$$P(B) = \frac{1}{3} \quad \underline{\text{Ans}}$$

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{array}{l} a \div b \\ a \times \frac{1}{b} \end{array}$$

Q#17: A fair die is thrown twice. Find the probability that an even number of dots appear in first and the number of dots in the second throw is less than 4.

Solution:

For a single die is thrown $= S = \{1, 2, 3, 4, 5, 6\}$
 $O(S) = 6$

when Even number of dots appear in first then

$$A = \{2, 4, 6\}$$

$$O(A) = 3$$

when less than 4 number of dots appear in second then:

$$B = \{1, 2, 3\}$$

$$O(B) = 3$$

Now, find Probability of event A and B:
 $P(A) = \frac{O(A)}{O(S)} = \frac{3}{6} = \frac{1}{2}$, $P(B) = \frac{O(B)}{O(S)} = \frac{3}{6} = \frac{1}{2}$

$P(A \cap B) = P(A) \cdot P(B)$ $\therefore$ Since A and B are independent events:
 $P(A \cap B) = \frac{1}{2} \cdot \frac{1}{2}$

$$P(A \cap B) = \frac{1}{4} \quad \underline{\underline{\text{Ans}}}$$

Q#18: Three missiles are fired at a target. If the probability of hitting target are 0.5, 0.3 and 0.6 respectively, and if the missiles are fired independently, what is the probability that all the missiles hit the target?

Solution:

Now, given that:

$$P(A) = 0.5, \quad P(B) = 0.3, \quad P(C) = 0.6$$

Since three events are occur independently then their Probability is,

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

$$P(A \cap B \cap C) = 0.5 \cdot 0.3 \cdot 0.6$$

$$P(A \cap B \cap C) = 0.09 \quad \underline{\underline{\text{Ans}}}$$